

E-Commerce Sales & Profit Analysis

Project Overview

This project focuses on analyzing an E-Commerce dataset containing approximately 9,994 customer order records. The dataset includes important business variables such as:

- Sales
- Profit
- Discount
- Quantity
- Product Category
- Sub-Category
- Customer Segment
- Shipping Details
- Regional Information
- Order and Ship Dates

The primary objective of this analysis was to understand business performance, identify profitable and loss-making areas, study customer purchasing behavior, and generate insights using data visualization techniques.

The project successfully combines:

- Data cleaning
- Exploratory Data Analysis (EDA)
- Statistical understanding
- Business intelligence
- Data visualization

to support strategic business decision-making.

Data Cleaning & Preparation

Before performing analysis, the dataset underwent preprocessing and validation.

Data Cleaning Steps Performed

- Converted **Order Date** and **Ship Date** into datetime format
- Extracted:
 - Order Month
 - Order Year
 - Day Name
- Checked for:
 - Missing values
 - Duplicate records
 - Incorrect data types

Key Findings

- Total Records: ~9,994
- Missing Values: 0%
- Duplicate Records: 0%

This confirmed that the dataset was clean, structured, and reliable for analysis.

Sales Performance Analysis

1. Overall Sales Performance

The business generated strong sales across multiple categories and customer segments.

Key Observation

Sales performance showed a consistent upward trend during the year, with noticeable peaks during festive and holiday seasons.

Business Interpretation

Higher sales during year-end months indicate:

- Increased customer demand
- Successful seasonal purchasing patterns
- Potential impact of promotions and discounts

This suggests that strategic marketing campaigns during these periods can significantly improve revenue generation.

2. Monthly Sales Trend Analysis

The monthly sales line chart revealed important seasonal patterns in customer purchasing behavior.

Major Findings

- Highest sales were recorded during:
 - November
 - December
 - September
- Lower sales were observed during:
 - January
 - February

Percentage-Based Insight

Approximately:

- 30–35% of annual sales were generated during the last quarter (Q4) of the year.
- December alone contributed a significant percentage of yearly revenue.

Business Insight

The company experiences strong seasonal demand during holiday and festive periods.

Recommendation

- Increase inventory during high-demand months
 - Launch aggressive marketing campaigns in Q4
 - Prepare logistics and supply chain operations in advance
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3. Category-Wise Sales Analysis

The category-wise pie chart showed the contribution of each product category to total sales.

Category	Approximate Contribution
Technology	~36%
Furniture	~32%
Office Supplies	~31%

Key Findings

- Technology contributed the highest share of total revenue.
- Furniture and Office Supplies contributed almost equally.

Business Interpretation

Technology products are highly preferred by customers and represent the strongest revenue-generating category.

Recommendation

- Continue investment in technology products
 - Expand premium and high-demand electronic product lines
 - Improve cross-selling strategies
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4. Sub-Category Sales Analysis

The sub-category sales visualization highlighted the best-selling products.

Highest Sales Generating Sub-Categories

- Phones
- Chairs
- Storage
- Tables
- Binders

Percentage Insight

The top 5 sub-categories contributed approximately:

- 45–50% of total sales revenue

Low Performing Sub-Categories

- Fasteners
- Labels
- Envelopes

These products contributed less than:

- 2–3% of overall sales

Business Insight

A small number of products generate the majority of sales revenue, indicating a strong concentration of customer demand.

Recommendation

- Focus inventory management on high-demand products
 - Review pricing and promotion strategies for low-performing products
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Profitability Analysis

5. Overall Profit Analysis

Profit analysis revealed that high sales do not always guarantee high profitability.

Key Findings

- Several categories generated strong revenue but lower profit margins.
 - Discounts and operational costs significantly affected profitability.
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6. Monthly Profit Trend Analysis

The monthly profit chart showed fluctuations across different months.

Major Findings

- Highest profits were generated during:
 - December
 - November
 - September
- Some months experienced reduced profit despite good sales performance.

Percentage Insight

Approximately:

- 35–40% of total annual profit was generated during Q4.

Business Interpretation

End-of-year purchasing not only increased sales but also improved profit generation.

Recommendation

- Focus on high-margin products during festive seasons
- Reduce unnecessary discounting
- Optimize operational costs

7. Profit Analysis by Category

Category	Profitability Performance
Technology	Highest Profit
Office Supplies	Stable Profit
Furniture	Lowest Profit Efficiency

Percentage Insight

Technology generated approximately:

- 50%+ of overall company profit

Furniture contributed a much smaller percentage despite strong sales volume.

Business Interpretation

Furniture products may involve:

- Higher shipping costs
- Higher production costs
- Excessive discounting

Recommendation

Review pricing and operational efficiency within Furniture products.

8. Sub-Category Profit Analysis

The profit distribution chart identified both highly profitable and loss-making products.

Most Profitable Sub-Categories

- Copiers
- Phones
- Accessories
- Paper

These products contributed a major share of overall profit.

Loss-Making Sub-Categories

- Tables
- Bookcases
- Supplies

Percentage Insight

Some sub-categories showed negative profit margins despite contributing positively to sales.

Business Interpretation

This indicates:

- High discounts
- Poor cost management
- Low profit margins

Recommendation

- Re-evaluate pricing strategy
 - Reduce discount dependency
 - Optimize logistics costs
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Customer Segment Analysis

9. Sales Distribution by Customer Segment

Segment	Approximate Sales Contribution
Consumer	~50%
Corporate	~30%
Home Office	~20%

Key Findings

- Consumer customers generated the largest share of revenue.
- Corporate customers maintained consistent purchasing behavior.
- Home Office customers contributed the smallest share.

Business Interpretation

The Consumer segment is the primary business driver.

Recommendation

- Prioritize customer retention strategies
- Offer loyalty programs

- Improve personalized marketing campaigns
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10. Profitability by Customer Segment

Major Findings

- Consumer segment generated the highest total profit.
- Home Office customers demonstrated better profit efficiency relative to sales.

Sales-to-Profit Ratio

Segment	Sales-to-Profit Ratio
Consumer	8.66
Corporate	7.68
Home Office	7.13

Interpretation

- Lower ratios indicate higher efficiency.
- Home Office customers generated more profit per unit of sales.

Business Recommendation

The company should:

- Improve profitability in Consumer sales
 - Focus on premium offerings
 - Reduce discount dependency
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Shipping & Operational Insights

The dataset also highlighted the importance of operational efficiency.

Key Findings

- Fast shipping methods may increase operational costs.
- Some products with high shipping costs showed low profitability.

Business Impact

Shipping and logistics directly influence net profit margins.

Recommendation

- Optimize delivery operations
 - Use cost-efficient shipping methods
 - Improve warehouse management
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Strategic Recommendations

Business Growth Recommendations

1. Focus on High-Profit Categories

Prioritize:

- Technology
 - Phones
 - Copiers
 - Accessories
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2. Improve Loss-Making Products

Re-evaluate:

- Tables
- Bookcases
- Supplies

by improving:

- Pricing strategy

- Cost management
 - Discount policies
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3. Strengthen Seasonal Campaigns

Since Q4 contributes nearly:

- 35–40% of yearly sales and profit,

the company should:

- Increase inventory
 - Improve logistics
 - Launch targeted promotions
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4. Improve Customer Retention

Focus on:

- Consumer customers
- Home Office customers

through:

- Loyalty programs
- Personalized recommendations
- Better customer engagement